

News & Social Media Analytics Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The **News & Social Media Analytics Engine** is a large-scale data processing platform designed to collect, organize, and analyze news articles and social media posts efficiently. The system uses **Data Structures and Algorithms (DSA)** to handle millions of content records, detect duplicate or highly similar articles, group related stories into meaningful clusters, and prioritize posts for moderation. Data structures such as **hash tables, graphs, priority queues, heaps, and trees** are used for efficient storage, searching, similarity checking, ranking, and moderator assignment. Hashing enables fast duplicate detection, while graph-based relationships and clustering algorithms help identify articles and posts discussing the same event or topic.

To support large-scale processing, the engine applies **parallel processing and efficient algorithmic techniques** so that multiple posts can be analyzed simultaneously instead of processing them sequentially. A priority queue can rank posts based on factors such as relevance, engagement, or moderation urgency, while a load-balancing strategy distributes moderation tasks among available moderators. The system therefore demonstrates practical applications of **searching, sorting, hashing, graph algorithms, clustering, priority queues, and parallel algorithms** in a real-world big-data environment. By combining these DSA concepts, the proposed engine aims to reduce duplicate information, organize related news efficiently, improve moderation workflow, and provide scalable analytics for millions of posts

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